Multiple Linear Regression & Multicollinearity

What is MLR?
It is a statistical techinque used to explain or predict one dependent variable using two or more independent variable.

Example:
Company predicting sales
- → expenditure
- → price
- → no. of salesperson

Dependent Variable: The outcome we want to explain or predict
Independent variable: The factor used to explain or predict the outcome.

Business problem	Dependent variable(Y)	Independent Variable(X)
House-price precting	House price	Size,bedrooms,age,
Sales Forcasting	Monthly sales	Ad Spend, price, sales team size
Employee attrition analysis	Attrition outcome	Salary, tenure, satisfaction
Delivery Time-analysis	Delivery Time	Distance, traffic, order size.

Simple vs Multiple linear regression

Simple Regression

$$y = a + bX$$

one predictor

Ex → predict sales from ad spend only

Multiple regression

$$y = a + b_1X_1 + b_2X_2 + \dots + b_kX_k$$

Two or more preditctors
Example: Predicting sales from ad spend, price, and store traffic, no of salesperson.

Simple Linear Regression considers one predictor at a time. Multiple regressions estimates how several predictors relate to the outcome within same model.

The Multiple Regression Formula

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + \ldots b_kX_k$$

$\hat{y}$ → predicted outcome

b_0 (ca) → Intercept or model's baseline

$b_1, b_2 \ldots$ → Regression coefficients

X_1, X_2 → Predictor variables

k → No. of predictors

Example: Predicting Monthly Sales

Suppose a company estimates the illustrative model"

$$\text{Sales} = 50 + 4A - 2P + 3R$$

A : Advertising spend in ₹10,000 units
P : price index units
R: number of sales represtatives

scale : ₹1,000 units

For a store with A= 5, P = 10 and R = 4

$$\hat{\text{Sales}} = 50 + 4*5 - 2*10 + 3*4 \qquad 62$$

Predicted sales = ₹62,000 under these illustrative units

Interpretation

The model predicts sales using tree factors together .The coefficients are hypthetical and do not establish that changing advertising or staffing will necessarily cause sales to change by those amounts.

Understanding Coefficients

Consider a equation

Price = 5.23 +0.045(Size) + 1.08 (Bedroom) - 0.17(Distance) - 0.08(Age)

Coefficients

Size: (+0.045)
Larger homes are associated with higher predicted prices, holding other included vaiables constant
Bedrooms: (+1.08)
One additional bedroom is associated with 1.08 unit increase in predicted price, holding the predictors constant
Distance: (-0.17)
One additional unit is associated with a 0.17 unit decrease in predicted price, all else equal
Age : (-0.08)
Each additional age unit is associated with 0.08 unit decrease in predicted price, all else equal.

"Holding other variables constant"
In multiple regression, a coefficient decribes the relationship between aq predictor and the predicted coutcome while other included predictors are held fixed.
This is important because larger house may also have more bedrooms. We need to distinguish the relationship associated with house size from the relationship associated with bedroom count

How do we calculate a regression Model?
Three Important concepts:
1. R^2 -- Coefficient of Determination
Measures the proportion of variation in the outcome explained by the fitted model, under the usual definition.
Ex: R^2 = 0.80 means the model explains 80% of the observed variation in the outcome in the dataset used fr calcualation. It does not mean the model is 80% accurate on every prediction.

2. Adjusted R^2
Adjust R^2 for the number of predictors and sample size. It helps when comparing models with different numbers or predictors, although it should not be used alone.

4. Residual
The difference between an observed value and it's predicted value:

$$e = Y - \hat{Y}$$

Example: Actual house price = ₹50 lakh; predicted price = ₹46 lakhs
Residual = 50-46 = 4 lakhs
The model underpredicts the observed price by ₹ 4 lakhs.

Residual depends on the context either it is good for the business or bad for the business, either it is positive or negative

Actual sales: 100 units
Predicted = 110
Residual = 100-110 = -10 units

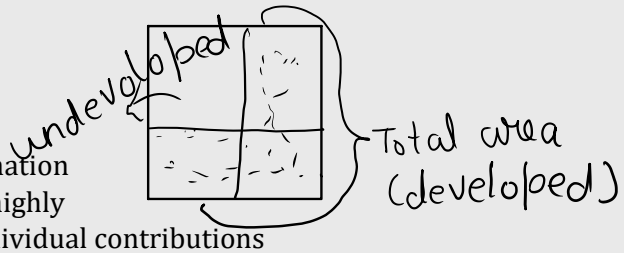
Why Multiple regression can be misleading

Example:

To predict house prices, the company includes;

1. House size in Square Feet
2. Total build-up area in sqaure feet.

These variables may contain overlapping information depending on how they are defined. If they are highly related, it becomes difficult to separate their individual contributions



Multicollinearity

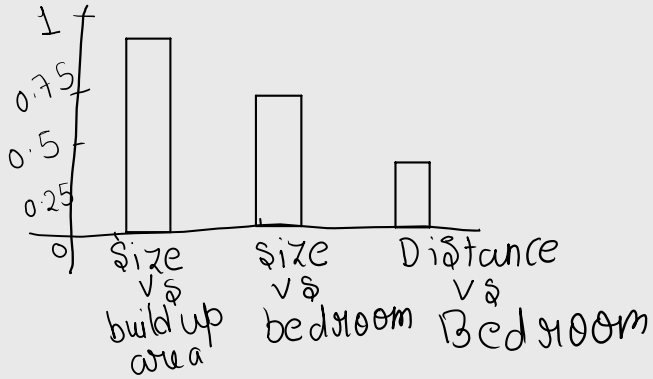
Multicollinearity occurs when two or more independent variables in a regression model are strongly related to one another.

It doesn't mean the outcome is strongly related to the predictors. The concern is the relationship between the predictors themselves.

Practical Example: Real Estate

X1: House Size
X2: Total buildup area
X3 : Number of Bedrooms
Y : House price

House size and total buildup area may be highly correlated. If both measure almost the same property characterisitic, the model may struggle to distinguish the individual coefficients.



The first pair has a very strong relationship in this example. Correlation alone does not establish whether a predictor should be removed.

Why Multicollinearity a problem?

1. Regression coefficients can become unstable
2. Coefficient standard error can increase
3. A predictor may appear statistically insignificant even when the predictors collectively help explain the outcome
4. Coefficient may change substantially when the data or model changes
5. It becomes harder to interpret each predictors separate contribution

Business Consequence
A real estate analyst might wrongly conclude that the house size in unimportant because coefficient is unstable after including a closely related area measure.

Detecting multicollinearity

Method 1: Correlation matrix
It shows pariwise relationship between numerical predictors
A high absolute correlation often around 0.8-0.9 -- No further investigation.

Remember:
Multicollinearity can involve a combination of several predictors when no individual pair has an extreme correlation.

Method 2: Variance Inflation Factor (VIF)
VIF measures how much the variance of regression coefficient is inflated because a predictor is related to the other preddictors.

$$VIF_j = \frac{1}{1 - R_j^2}$$

Suppose the auxilliary regression for the house size gives $R_j^2 = 0.9$

$$VIF = 1/(1-0.90) = 1/0.10 = 10$$

So our VIF = 10
A VIF of 10 indicates substantial multicollinearity may be present.
Common investigation threshold includes 5 and 10 but these are just guideline, not absolute laws

Method 3: Tolerance
Tolerance is reciprocal of VIF

$$\text{Tolerance} = 1 - R_j^2 = 1/VIF$$

For same example:

$$\text{Tolerance} = 1/10 = 0.10$$

Low tolerance indicates that much of predictor's variation can be explained by other predictors.

VIF (10) : Investigates high multicollinearity
Tolerance (0.10) : Predictor has substantial overlap with others.

How should an analyst handle multicollinearity

1. Check the preditcors
Review the definitions, correlation matrix, VIF and tolerance
2. Understand the business Meaning
Are two variables measuring nearly the same thing, or are both genuinely important?
3. Choose a suitable response
Remove redundant varaible if justified, combine measures meaningfully, collect better data, or consider a regularized model when appropriate.

4. Refit and evaluate

Check the coefficient stability, interpretability and predictive performance on suitable validation data

Important Point:

Never remove a variable just because its correlation is high. Check the business meaning, model purpose, and effect of the change.

House Size	bedrooms	Distance	Age	House price
2000	3	0.5	4	52
1500	3	0.4	2	46